

Conductive bacterial cellulose/multiwall carbon nanotubes nanocomposite aerogel as a potentially flexible lightweight strain sensor

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ABSTRACT

In this work, in-situ biosynthesized bacterial cellulose (BC) /multiwall carbon nanotubes (MWCNTs) nanocomposite hydrogels converted to the conductive nanocomposite aerogels via the supercritical CO₂ method. A low percolation threshold value of 0.0041 (volume concentration) predicted for BC/MWCNTs nanocomposite aerogels by the proposed modified model. The piezoresistive behavior of the nanocomposite aerogel at percolation threshold, evaluated in tension mode. The strain sensing outcomes revealed a linear trend during loading until a critical strain, afterward began to decline with further increasing of strain. Moreover, by applying loading unloading cyclic tension for 10 times at two different strain amplitudes (2% and 8%), the variation of relative resistance was different. This attributed to the rearrangement of MWCNTs at high strain condition. The gauge factor of 21 and response time of 390 ms obtained for flexible lightweight strain sensor. The fabricated strain sensor utilized to monitor human detection motion.

1. Introduction

Cellulose aerogels as a third generation of aerogels have attracted a great deal of attention in energy storage devices (Zhang, Zhang, Zhao, & Yang, 2015), water purification (Wan & Li, 2015) and smart magnetic materials (Olsson et al., 2010), due to their high porosity (> 80%), low density (0.004–0.5 g/cm³), flexibility, high specific surface area and three-dimensional interconnected fibrous structure (Demilecamps, Beauger, Hildenbrand, Rigacci, & Budtova, 2015).

Generally, resistance-type strain sensors, which convert the external stimuli (stress or strain) to electrical resistance signal, have been extensively used in the electronic skin (Minjeong et al., 2015) and wearable devices (Liu, Cao, Ma, & Wan, 2017). Nevertheless, pivotal requirements for the strain sensing material are including highly flexible matrix and efficient conductive nanofiller network. Recently, the utilization of cellulose, as one of the most abundant biopolymers on earth, reported in the field of strain sensing devices. For example, Huang, Liu, Wu, Li, and Wang (2017) fabricated composite aerogels based on graphene/carboxymethylcellulose for compressive strain sensing evaluation and a gauge factor (GF) value of 1.58 obtained (Huang et al., 2017). Yao et al. (2017) reported flexible Ag/cellulose nanofiber aerogel, gained from bamboo, with a maximum GF of 1 up to 20% strain (Yao et al., 2017). Zhuo et al. (2018) introduced a carbon aerogel via carbonization of cellulose nanocrystalline/graphene oxide

in compression strain and pressure detection with a GF of 5.81 (Zhuo et al., 2018).

It explains that the presence of cellulose plays a critical influence on the viscoelastic properties of the resultant strain sensor. Moreover, the homogeneous dispersion of carbon-based nanofiller within the polymer matrix is vital to improve the performance of sensory materials.

BC possesses high purity, crystallinity, and higher flexibility than plant cellulose or other derivatives of cellulose (Ul-Islam, Khan, & Park, 2012). A variety of conductive nanofillers such as carbon nanotube (CNT) and graphene introduced into BC aerogels. Accordingly, the in-situ method suggested as a proper strategy for achieving conductive BC based nanocomposite due to adjusting the shape, structure, and properties of resultant BC along with excellent dispersion of nanofillers during cultivation and biosynthesis process (Erbaş Kiziltas, Kiziltas, Blumentritt, & Gardner, 2015). A few works have been reported about the incorporation of MWCNTs into the BC culture medium and characterize resultant composites (Park, Kim, Kwon, Hong, & Jin, 2009; Yan, Chen, Wang, Wang, Wang et al., 2008; Yan, Chen, Wang, Wang, & Jiang, 2008), or evaluate bone regeneration (Park et al., 2015) and enzymatic biofuel cell (Lv et al., 2016). No one investigated the electrical conductivity and strain sensing behavior of composites.

The literature survey clearly illustrates lack of experimental and theoretical works on the electrical conductivity and strain sensing behavior of in-situ biosynthesized BC/MWCNTs nanocomposite aerogels.

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Furthermore, the designing along with the development of flexible, lightweight, and conductive nanocomposite aerogels as well as prediction of their overall electrical conductivity is immensely challenging. It has been expressed that the variations of resistance for strain sensors around the percolation threshold are significantly sensitive to tensile strain (Hu et al., 2017). Hence, the calculation of percolation threshold is the crucial step for evaluating the strain sensing behavior. In our previous work (Hosseini, Kokabi, & Mousavi, 2018), a model modified by the implementation of porosity as a third term and proposed for predicting the percolation threshold and electrical conductivity of BC/rGO nanocomposite aerogels. The effective medium approach that pointed out by Wang, Weng, Meguid, and Hamouda (2014) utilized as a basic model, owing to its computational simplicity, determination of percolation threshold, dealing with the electron tunneling along with interface effect, and also compatibility with our physical case. Herein, the proposed model applied for predicting the percolation threshold and overall electrical conductivity of BC/MWCNTs nanocomposite aerogel.

In this study, based on predicted percolation threshold value, various contents of MWCNTs, around predicted value, incorporated into the culture medium solution. Subsequently, *G. xylinus* bacteria inoculated into culture media, and the nanocomposite hydrogels harvested at the end of the culture period. The obtained nanocomposite hydrogels then converted to nanocomposite aerogels via ScCO_2 drying method to obtain electrical conductive nanocomposite aerogels. The products characterized and compared with neat BC aerogel using FTIR, N_2 adsorption-desorption methods, and FESEM analysis. Finally, the strain sensing behavior of conductive nanocomposite aerogel evaluated in the tension mode.

A theoretical approach for prediction of percolation threshold and overall electrical conductivity presented as the Supplementary Information.

2. Material and methods

2.1. In-situ biosynthesis of BC/MWCNT hydrogels and fabrication of nanocomposite aerogels

BC/MWCNTs nanocomposite hydrogels prepared using the in-situ biosynthesis method similar to our previously reported work for BC/rGO (Hosseini et al., 2018). According to the theoretical prediction of percolation threshold value, 0.5, 0.7, 0.9, 1.1 wt% (based on culture medium solid content) of acid-treated MWCNTs (purchased from Research Institute of Petroleum Industry of Iran) considered and added to the culture medium solutions and autoclaved. Each culture medium solution contained (2% (w/v) glucose, 0.5% (w/v) yeast, 0.5% (w/v) peptone, 0.115% (w/v) citric acid, and 0.27% (w/v) $\text{Na}_2\text{HPO}_4 \cdot 12\text{H}_2\text{O}$) and adjusted on a pH of 6.5. All the culture medium materials acquired from Merck, Germany. Seed culture broth with *Gluconacetobacter xylinus* (PTCC 1734, purchased from the Iranian Research Organization for Science and Technology (IROST)) inoculated to the latter culture mediums and transferred to a rotary shaker incubator operating at a rotational speed of 100 rpm at 28 °C for 6 days to prevent the settling of MWCNTs during incubation process. Finally, the biosynthesized nanocomposite hydrogels washed with 1% (w/v) sodium hydroxide (NaOH) solution for 1 h at 85 °C to remove bacterial cell debris and then purified and cleansed using distilled water several times to reach a neutralize pH. The harvested products converted to the aerogels by the ScCO_2 drying method at a temperature and pressure of 40 °C and 100 bar, respectively. The details described in Supplementary Information.

2.2. Products characterizations

Evaluating the morphology and microstructure of the fabricated aerogels a TESCAN MIRA 3-XMU (Republic of Czech) field-emission scanning electron microscopy (FE-SEM) employed. Before observation,

the samples mounted on a metal stub and sputter coated using a gold sputtering machine with a layer of gold approximately 100 Å thick to reduce charge interruptions.

The DC electrical conductivities of nanocomposite aerogels investigated using a broadband dielectric spectrometer, GW Instek, LCR-8101 G LCR meter-Taiwan. Before testing, a silver paint rubbed over the two surfaces of each specimen to diminish the contact resistance between sample and probes.

The strain sensing capability of BC/MWCNTs nanocomposite aerogel, at the percolation threshold, examined by subjecting to a tensile test via Gotech universal testing machine (GT-TCS 2000, Taiwan). Simultaneously, the variation of resistivity recorded by a digital multimeter (Agilent 34401A). A constant crosshead speed of 1 mm/min applied for loading unloading. The reproducibility of strain sensor evaluated at two different strain amplitudes (2% and 8%) upon 10 cycles. Thrice replicates carried out for each experiment and representative findings recorded. The experiment setup for a strain sensor testing presented in Fig. S1. The flexibility of fabricated strain sensor evaluated over 100 cycles of loading unloading in twisting mode. Accordingly, the variation of resistance recorded using a digital multimeter (Agilent 34401A) upon 100 and 1000 continuous cycles in twisting and bending modes, respectively. These experiments carried out thrice for each of three samples and comprehensive results rendered.

3. Results and discussion

3.1. Morphology and microstructure of aerogels

Fig. 1(A) and (B) represent the FESEM micrographs of BC aerogel. The BC aerogel architecture consists of an interconnected highly porous network with randomly distributed nanofibers. All nanofibers have disorder assembly. The mean diameter of BC nanofiber estimated at 40 nm. The noticeable feature evident in the FESEM micrograph of as-received MWCNTs indicates the presence of partially U-shaped nanotubes that entangle to each other and randomly orient as a bundle, Fig. 1(C). This attributes to the flexibility of MWCNTs, resulting in a U-shaped morphology that remarkably enhances the chance of elastic interlocking of MWCNTs (Sadeghi, Arjmand, Otero Navas, Zehtab Yazdi, & Sundararaj, 2017). The BC growth rout in the presence of MWCNTs presented in Fig. S2.

Fig. 1(D–F) depict the FESEM micrographs of the in-situ biosynthesized BC/MWCNTs nanocomposite aerogel at the percolation threshold. The BC porous and fibrous structure associated with the 3D interconnected network preserves in BC/MWCNTs nanocomposite aerogels.

After incorporation of MWCNTs into the BC culture media and in-situ biosynthesis process, the BC nanofibers and MWCNTs intertwine together and MWCNTs partially cover by BC nanofibers. It reported by Park et al. (2015). In addition, MWCNTs adhere strongly on the surface of BC nanofibers. This attributes to the interaction between the BC functional groups with the oxygen-containing groups of the carboxylic of MWCNTs. Moreover, a bright dots regions are clear in the FESEM images of BC/MWCNTs nanocomposite aerogel due to the high conductivity of MWCNTs, as they distribute uniformly in the BC structure without aggregation (Yoon, Jin, Kook, & Pyun, 2006). Thus, according to FESEM micrographs, the dispersion state of nanotubes is adequate in the 3D framework of BC.

3.2. Electrical conductivity measurements

The modeling and experimental results of the overall electrical conductivity as a function of the MWCNTs volume concentration for nanocomposite aerogels have presented in Fig. 2. The electrical conductivity data depicts a dramatic increase as the MWCNTs contents rise to a critical value, i.e. percolation threshold, suggesting the formation of MWCNTs continuous pathways in the aerogel nanocomposites.

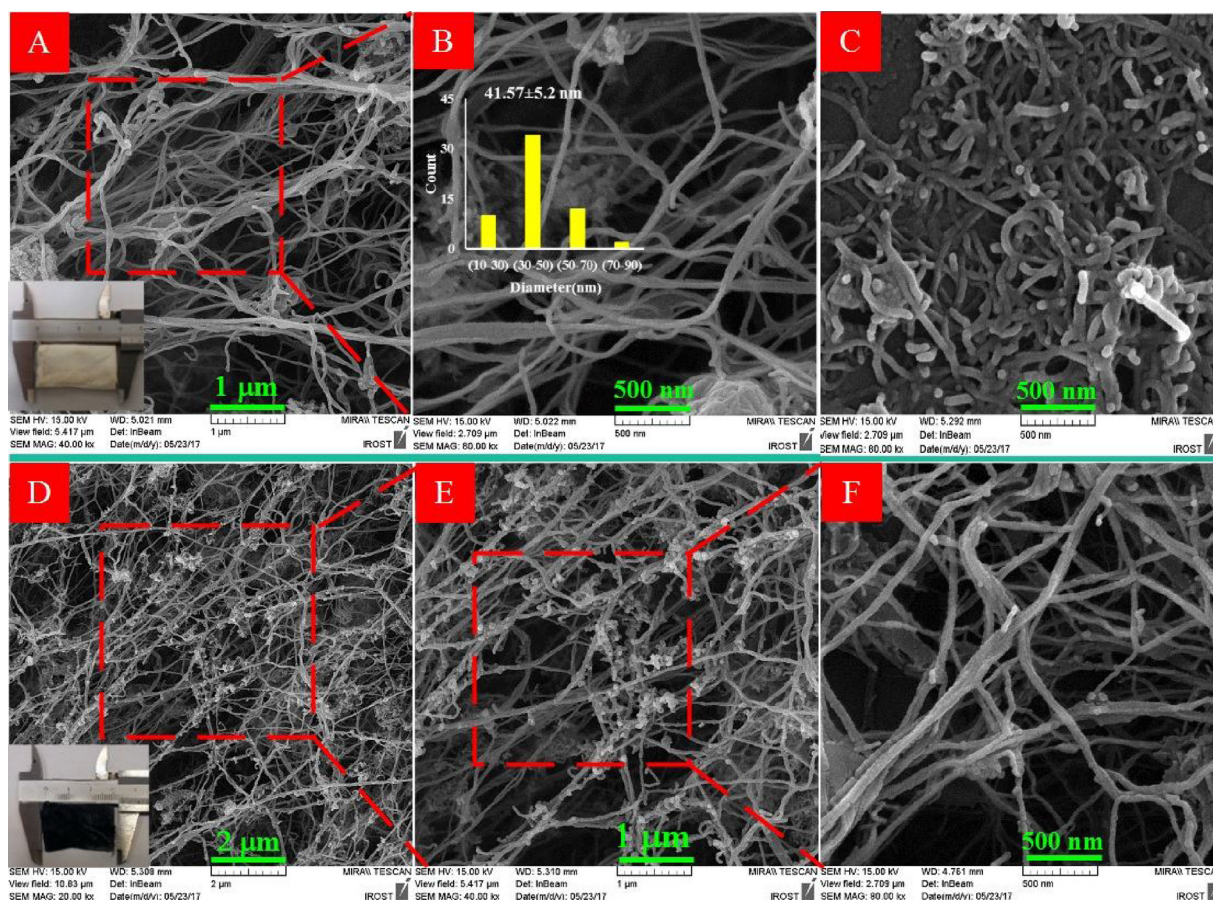


Fig. 1. FESEM micrographs of (A & B) BC aerogel, (C) as-received MWCNTs and (D–F) BC/MWCNTs nanocomposite aerogel at the percolation threshold with different magnifications.

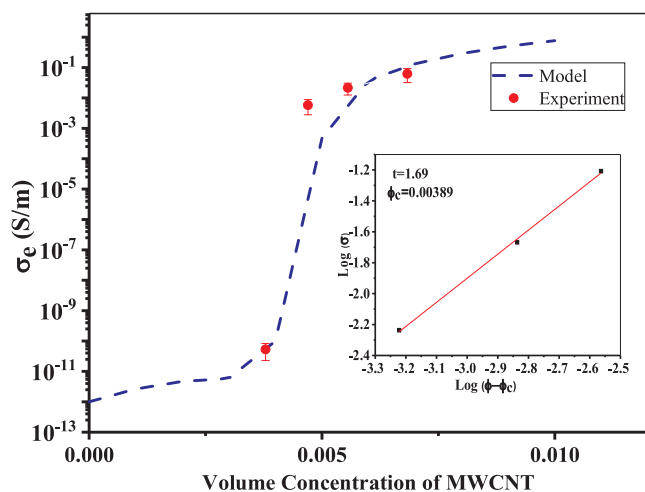


Fig. 2. The plot of predicted effective electrical conductivity compared with experimental data for BC/MWCNT nanocomposite aerogels (the inset illustrates the log conductivity against $\log(\phi - \phi_c)$).

Therefore, a transition from insulator nature to conductor state has occurred. Generally, at the low content of MWCNTs, beneath the percolation threshold, they have considered as isolated nanofillers, thereby leading to lack of conductive network formation. Further increasing of the MWCNTs content, leads to a percolated state, and the conductive network has created throughout the nanocomposite aerogels. As a result, the electron can pass through the conductive network, causes a metal-like conduction behavior (Sun et al., 2017). The experimental

findings show that the electrical conductivity quickly rises to 10^{-2} S/m at the MWCNTs content of 0.0041 (volume concentration). This value, for MWCNTs content, corresponds to the percolation threshold, which has predicted by the proposed modified model (Hosseini et al., 2018). However, experimental data can fit a power law functionsuch as Eq. (1) to obtain precise percolation threshold value:

$$\sigma = \sigma_0(\phi - \phi_c)^t \quad (1)$$

Where σ , σ_0 , ϕ_c and t are the conductivity of the sample, constant parameter, percolation threshold and critical exponent, respectively. A percolation threshold of 0.00389 has estimated from the fitting outcomes (the inset in Fig. 2). Higher percolation threshold value observed for BC/rGO nanocomposite aerogels (0.0045 vol concentration) in our previous work (Hosseini et al., 2018). The lower percolation threshold value in this work attributes to the interlocking ability of carbon nanotubes. Nevertheless, a small discrepancy detects between predicted percolations thresholds with the experimental value. It relates to the surface energy between BC with MWCNTs which cannot adopt in the Eq. S(3). We are of the opinion that the precise calculation of the carbon nanotube dimensions followed by aspect ratio after dispersion is very complicated which has a remarkable influence on the prediction of percolation threshold. However, the DLS method presents a convenient approximation of carbon nanotubes dimensions after full dispersion in the culture medium solution.

It clearly shows the theoretical curve agrees with the experimental data. This outcome confirms the efficacy of the modified model. As mentioned, the effect of porosity is negligible beyond the percolation threshold due to the formation of the conductive network associated with electron tunneling phenomenon. Although, the introduction of porosity leads to the drop in overall electrical conductivity beneath the

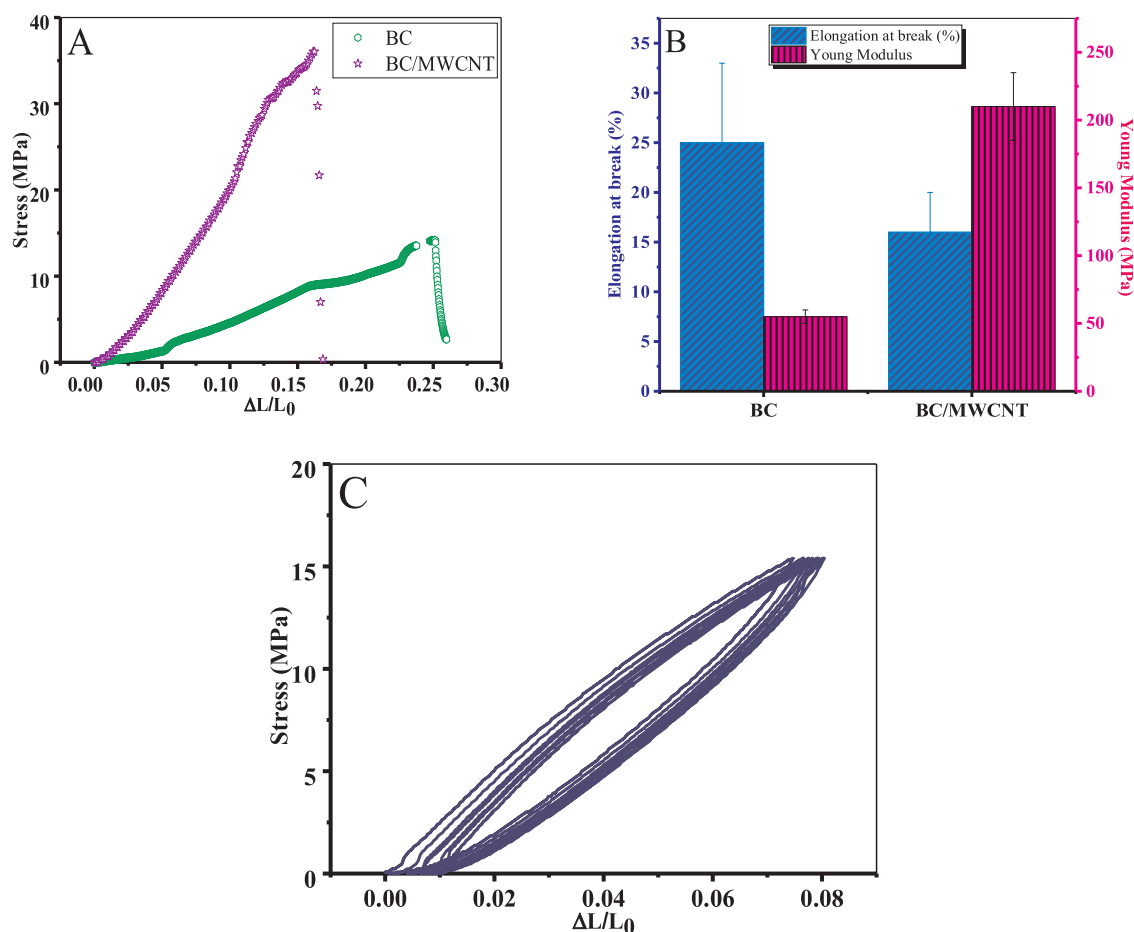


Fig. 3. (A) The typical stress-strain curves and (B) Specific mechanical properties of BC neat aerogel and BC/MWCNTs nanocomposite aerogel at the percolation threshold (0.41%vol), (C) Relationship of stress against strain for BC/MWCNTs nanocomposite aerogel at the percolation threshold (0.41%vol) over 10 loading-unloading cycles (in tension mode).

percolation threshold. The observation of close agreement between experimental data with the prediction of the modified model confirms our claim about the effect of porosity on the overall electrical conductivity of nanocomposite aerogels. Similar results reported for electrical conductivity of polycarbonate/carbon nanotube composite foam (Zhi et al., 2015). It showed that the electrical conductivity of composite foam, beyond percolation threshold, was identical with bulk condensed composite. This observation attributed to preserving of the carbon nanotubes conductive network in the presence of insulating pores. Thus, the pores cannot disrupt the conductive network of nanotubes. However, the introduction of the pores may affect the uniform distribution of the conductive network (He, Zhang, & Xu, 2012). We observe that nanocomposite aerogel behaves the same as matrix beneath the percolation threshold; thereby nanocomposite aerogel exhibits insulating nature. However, based on the modeling results, the introduction of 0.41 vol% of nanotubes into the BC framework is adequate to gain a conductive nanocomposite aerogel.

3.3. Mechanical properties of neat and nanocomposite aerogels

Fig. 3(A) represents the typical stress-strain curves of BC aerogel and BC/MWCNTs nanocomposite aerogel at the percolation threshold. By incorporation of MWCNTs into BC structure the elongation at break of BC/MWCNTs nanocomposite aerogel decreases (16%), although, its Young's modulus increases up to 312 MPa in comparison with neat BC aerogel (Fig. 3(B)). This attributes to the good dispersion associated with the proper interaction between carbon nanotubes and the BC nanofibers matrix. Another reason for this observation arising from high

modulus and strength of carbon nanotube that endows it an excellent candidate for improvement of mechanical properties.

In addition, Fig. 3(C) shows the mechanical properties under cyclic tension over 10 cycles up to a maximum strain of 8% for BC/MWCNTs nanocomposite aerogel. In each cycle by increasing of strain to a maximum value, the stress rises linearly. When the strain attains the maximum value, the load removes and the stress comes back to the original value. Upon these processes, the hysteresis behavior, which originates from the viscoelastic effect and deformation, observes. The similar observation has been expressed for PLA/CNT (Hu et al., 2017) and PP/CNT (Zhao et al., 2013) nanocomposite. It is clear that by reaching the stress into zero value, the strain does not retrieve to the original point. In addition, the maximum value of stress declines upon an increment of cycle number that attributes to the viscoelastic nature of BC matrix. Accordingly, the BC chains possess no sufficient time for relaxation process by the increment of the cycle number.

3.4. Strain sensing behavior

Fig. 4(A) depicts the variation of relative resistivity (the immediate variation of resistivity, ΔR , per initial resistivity, R_0) along with stress versus strain. Stress rises linearly with strain up to specimen failure. However, for $\Delta R/R_0$ against the strain, the change of relative resistivity increases linearly with an increment of strain and subsequently slightly declines at critical strain until the failure of the specimen. The linear increase of $\Delta R/R_0$ vs. strain corresponds to the breakdown of the conductive network during elongation. This is resulting from the increment of the separation distance between adjacent carbon nanotubes and

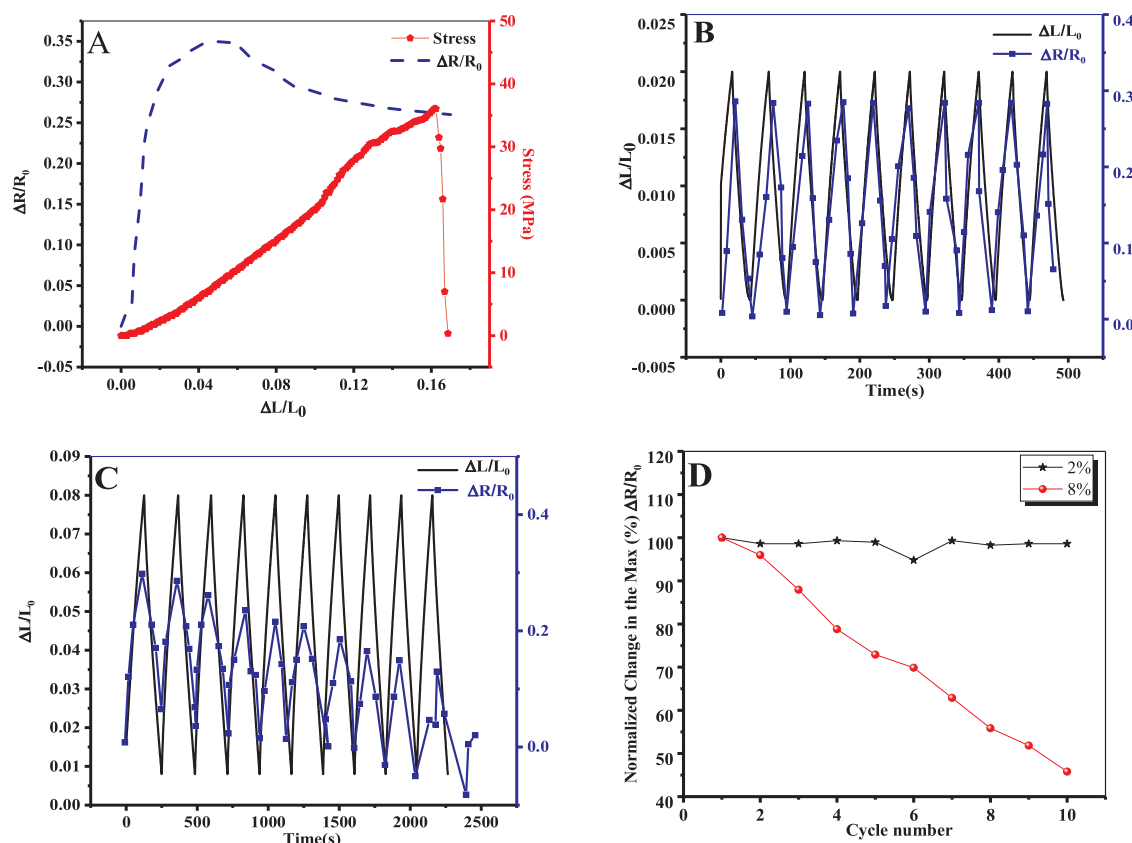


Fig. 4. (A) The variation of relative electrical resistivity (left axis) and tensile stress (right axis) versus strain, Cycling tensile loading of a BC/MWCNTs nanocomposite aerogel during 10 loading-unloading cycles under strain up to (B) 2%, and (C) 8%, (D) the variation of the max value $\Delta R/R_0$ for fabricated strain sensor with different tensile strain amplitudes over 10 cycles.

tunneling distance when the specimen subjects to a tension. In this region, the destruction of the nanocomposite aerogel and conductive network simultaneously takes place. By further increasing of strain, a steady region of $\Delta R/R_0$ can observe. It relates to the high aspect ratio and interlocking of carbon nanotubes, leading to the balance of destruction and reconstruction of conductive networks. In this region, the Poisson's contraction creates in the sample due to the occurrence of inelastic deformation in the nanocomposite aerogel, resulting in the reorientation and rearrangement of carbon nanotubes. At the higher applied strain and beyond the critical strain, the reduction of $\Delta R/R_0$ vs. strain is substantial. This behavior arises from the sever Poisson's effect, leading to closing nanotubes to each other and formation of new percolated paths. Therefore, a gradual drop of resistivity observes. The similar results reported for CNT based conductive composites (Cao et al., 2017). However, a different trend observed for carbon black (Zhao et al., 2013) and graphene (Liu et al., 2016) based strain sensors. This discrepancy attributes to the different conductive networks influenced by the type of nanoparticles.

Furthermore, the BC/MWCNTs nanocomposite aerogel exhibits a good strain sensitivity. The gauge factor (GF) of strain sensor is calculated using following formula:

$$GF = \frac{\frac{\Delta R}{R_0}}{\frac{\Delta L}{L_0}} \quad (2)$$

The gauge factor (sensitivity) of present novel conductive nanocomposite aerogel obtains 21. Table 1 illustrates the strain sensing performance of some conductive composites reported previously.

According to Table 1, the gauge factor value of present work is higher than that reported for other conductive nanocomposite systems, that arises from flexible and nanofibrous nature of BC that affecting the

piezoresistive nature of nanocomposite aerogel. In comparison with high stretchable elastomeric matrix strain sensors, a higher GF value obtained for BC/MWCNTs nanocomposite aerogel upon lower maximum strain. This outcome indicates the superiority of BC/MWCNTs nanocomposite aerogels as a strain sensor.

The variation of relative resistance against time at two different strain amplitudes (2% and 8%) over ten cycles was also measured with a view to evaluating reproducibility along with obtaining further insights about the microstructure of the resultant BC/MWCNTs nanocomposite aerogel strain sensor.

In Fig. 4(B) by applying the strain, the $\Delta R/R_0$ rises and subsequently reduces upon unloading. As stated earlier, in this case, the interparticle distances or tunneling distances between adjacent nanotubes increase, leading to the increment of resistivity. On the contrary, upon unloading, due to the reconstruction of new conductive networks, the decreasing of the relative resistance observes. These behaviors are more significant at higher strain amplitude (8%). In other words, at higher strain amplitude (8%) the max and min $\Delta R/R_0$ cannot attain their original values after the first cycle (Fig. 4(C)). This is due to hysteresis effect of specimen arises from irreversible deformation of nanocomposite aerogel which affects on the destruction and reconstruction of conductive networks. Accordingly, by further increasing the cycle number, the max and min values of $\Delta R/R_0$ dwindle. Generally, nanotube has a large aspect ratio and interlocking structure (according to FESEM micrographs in Fig. 1(C)), that preserves its conductive network in high strain ranges.

Thus, at high strain amplitude that resulting in the observation of hysteresis effect in the nanocomposite aerogel, a rearrangement of nanotubes creates and leads to the formation of the convenient conductive network during loading after every cycle. Consequently, the max and min values for $\Delta R/R_0$ depict a dramatic fall in the tenth cycle.

Table 1

Performance of the different stain sensors based on conductive nanocomposite.

System	GF	Content of nanofiller (wt%)	Strain rate (mm/min)	Maximum strain (%)	Reference
Vinylester/MWCNTs	2.6	1	1	1.5	Ku-Herrera, Avilés, and Seidel, (2013)
Epoxy/MWCNTs	0.66	0.3	0.5	12	Cao et al. (2017)
Cellulose/Ag	1	8.6	1	20	Yao et al. (2017)
Graphene/carboxymethylcellulose	1.58	10	–	70	Huang et al. (2017)
TPU/graphene	0.78	0.6	0.1	30	Liu et al. (2016)
TPU/MWCNTs	6.8	0.3	–	60	Zheng et al. (2017)
BC/MWCNTs	21	0.7	1	16	Present work

The variations of $\max \Delta R/R_0$ as a normalized parameter (the $\max \Delta R/R_0$ at each cycle divided by $\max \Delta R/R_0$ at first cycle) for above-mentioned strain amplitudes (2% and 8%) are displayed in Fig. 4(D). As observed, for the strain amplitude of 2%, the variations of the $\max \Delta R/R_0$ quantities are not substantial. This attributes to the linear elasticity of nanocomposite aerogel at this strain. It means, upon loading unloading, conductive networks return to the original structure. Although, for the strain amplitude of 8%, a significant drop in the $\max$ quantities of $\Delta R/R_0$ can be observed after every cycle. As mentioned before, this ascribes the rearrangement of nanotubes and causes the reconstruction of the further conductive pathways. A schematic diagram presented in Fig. S7 to better understanding the sensing behavior of resultant BC/MWCNTs nanocomposite aerogel.

The flexibility of the fabricated strain sensor depicted in Fig. 5(A) and (B) in twisting and bending mode, respectively. The BC/MWCNTs nanocomposite aerogel under the twisting condition for 100 cycles and over 1000 cycles in bending state exhibits an excellent flexibility and dimensional stability.

The relative resistance (R/R_0) tolerance, (R is the resistance after loading and, R_0 is the initial value for the resistance of nanocomposite aerogel), is around 5%. Moreover, in bending mode, the tolerance of (R/R_0) is nearly 4.5%. These relatively low values, confirm the stable and strong conductive network of carbon nanotubes in the BC framework that does not destruct upon harsh conditions, specifically in bending mode. It also indicates the flexibility and dimensional stability of BC/MWCNTs nanocomposite aerogel.

To further investigating the practical application of BC/MWCNTs nanocomposite aerogel strain sensor, it was attached to a latex glove to monitor the finger motions as depicted in Fig. 5(C). The relative resistance increases upon bending of the finger owing to the separation distance increase between adjacent nanotubes. Afterwards, the sensor reverts and the finger straightens, which causes the decrease in resistance due to decreasing of strain. This finding clearly proves the capability of the fabricated strain sensor in human motion detections.

Besides gauge factor, another vital parameter in the sensor-based material is the response time. The response time of strain sensor

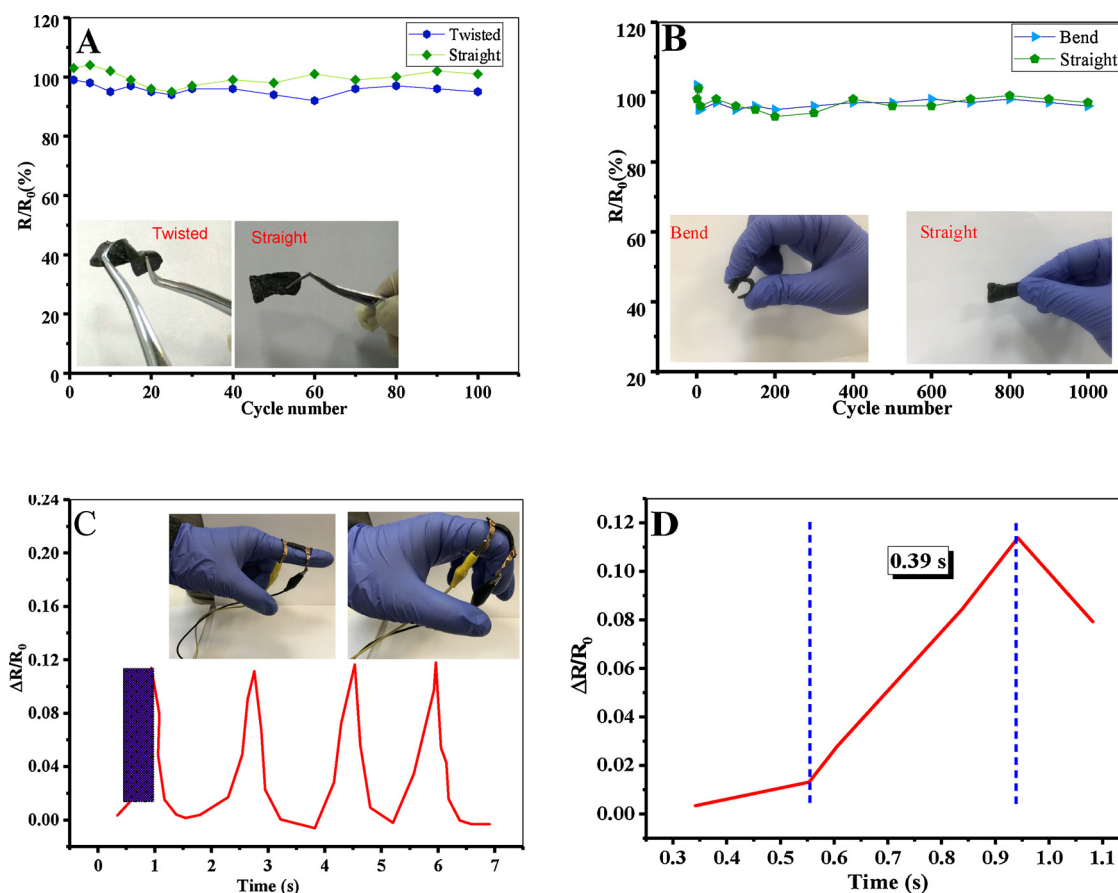


Fig. 5. (A & B) The flexibility test : The variation of resistance (A) over 100 cycles in twisting mode and (B) over 1000 cycles in tension mode (inset: photographs of loading unloading states of BC/MWCNTs nanocomposite aerogel), (C) The relative resistivity variation during finger bending, and (D) Response time of fabricated sensor.

indicates the velocity of mechanical stimuli. Therefore, a smaller response time gives rise to the observation of mechanical stimuli at the higher velocity. As shown in Fig. 5(D), the response time of fabricated sensor is about 390 ms that is comparable with that of reported for paper-based strain sensor (240 ms) (Liu, Jiang et al., 2017) and faster than previously reported strain sensors (400–3000 ms) (Ramuz, Tee, Tok, & Bao, 2012; Wang et al., 2013).

In order to expand on the clarifying the mechanism of fabricated strain sensor a mathematical modeling, based on tunneling theory, utilized. The resistance of conductive nanocomposites expressed by following equations (Liu et al., 2016):

$$R = \left(\frac{l}{N}\right) \left(\frac{8\pi h \zeta}{3\gamma a^2 e^2}\right) \exp(\gamma \zeta) \quad (3)$$

and:

$$\gamma = \frac{4\pi\sqrt{2m\psi}}{h} \quad (4)$$

Where l is the number of particles forming a single conductive pathway, N is the number of conductive pathways, h is the Planck's constant, ζ is the minimum distance between neighbouring conductive particles, a^2 is the effective cross-section area, e is the electron charge, m is the electron mass and ψ is the height of potential barrier between neighbouring particles.

By applying an extensional stress to the sample, an alteration of resistance (as a decline) observes due to the increment of separation distances between particles. This variation occurs linearly and correlates with increasing the tensile strain (ϵ) from ζ_0 to ζ . Therefore:

$$\zeta = \zeta_0 (1 + M\epsilon) \quad (5)$$

Where M is a constant value that is variable in different nanoparticle systems. On the other hand, the number of conductive pathways changes upon tensile loading according to Eq. (6):

$$N = \frac{N_0}{\exp(A\epsilon + B\epsilon^2 + C\epsilon^3 + D\epsilon^4)} \quad (6)$$

Where A–D are equation constants.

Thus, by substituting Eqs. (5) and (6) into Eq. (3), the normalized variation of resistance ($\Delta R/R_0$) defines as:

$$\frac{\Delta R}{R_0} = \frac{R - R_0}{R_0} = (1 + M\epsilon) \exp[(A + EM)\epsilon + B\epsilon^2 + C\epsilon^3 + D\epsilon^4] - 1 \quad (7)$$

where the constant of E is equal to $\gamma\zeta_0$.

In Fig. 6(A), a relatively good matching observed between the modeling results (based on Eq. (7)) with experimental findings. The

Table 2

Calculated parameters via fitting $\Delta R/R_0$ -strain curve using Eq. (6) for BC/MWCNTs nanocomposite aerogel.

	A	B	C	D	E	M
BC/MWCNT	0.50	−0.3347	0.243625	6.058×10^{-4}	125.6	0.09695

fittings parameters summarized in Table 2. The variation of the conductive pathway (CP) along with that of tunneling distance (TD) determined and presented versus applying strain (Fig. 6(B)).

It observed that the number of conductive pathways reduced significantly in the linear region. This suggests that conductive networks of nanocomposite aerogels will destroy under strain. On the other hand, tunneling distance rises linearly with increasing the strain. These findings confirm the strain sensing behavior of nanocomposite aerogel.

These results imply that prepared conductive MWCNT-based BC nanocomposite aerogel with a proper gauge factor has a great application potential as a flexible piezoresistive strain sensor. Our study opens a new door toward designing and producing flexible electronic devices and health monitoring sensors.

4. Conclusions

Flexible lightweight conductive BC/MWCNTs nanocomposite aerogels successfully in-situ biosynthesized followed by ScCO_2 drying method. The influence of carbon nanotubes on morphology and microstructure of BC nanofibers carefully investigated. The results indicated that the introducing of nanotubes into culture media, and subsequently fabricated BC, decreased average pore size and volume shrinkage of aerogels to 8.6 nm and 2%, respectively. Additionally, the specific surface area and bulk density of nanocomposite aerogel soared to $235 \text{ m}^2/\text{g}$ and $0.024 \text{ g}/\text{cm}^3$, respectively. Moreover, around predicted percolation threshold value, four different concentrations of MWCNTs incorporated into the BC culture medium. The results of the modified model showed a decline of overall electrical conductivity in the presence of porosity beneath the percolation threshold. This attributed to the lack of any conductive network along with electron tunneling effect. Meanwhile, a good agreement observed between model prediction and experimental data. Furthermore, strain-sensing behavior of fabricated nanocomposite aerogel under tensile loading evaluated. By increasing the strain, the relative resistivity increased and then slightly decreased at the critical strain. The strain sensing behavior over 10 cyclic loading/unloading depicted that the max value of $\Delta R/R_0$ at strain amplitude of 2% was not substantial. On the contrary, a dramatic fall of the max $\Delta R/R_0$ value observed for strain amplitude of 8%,

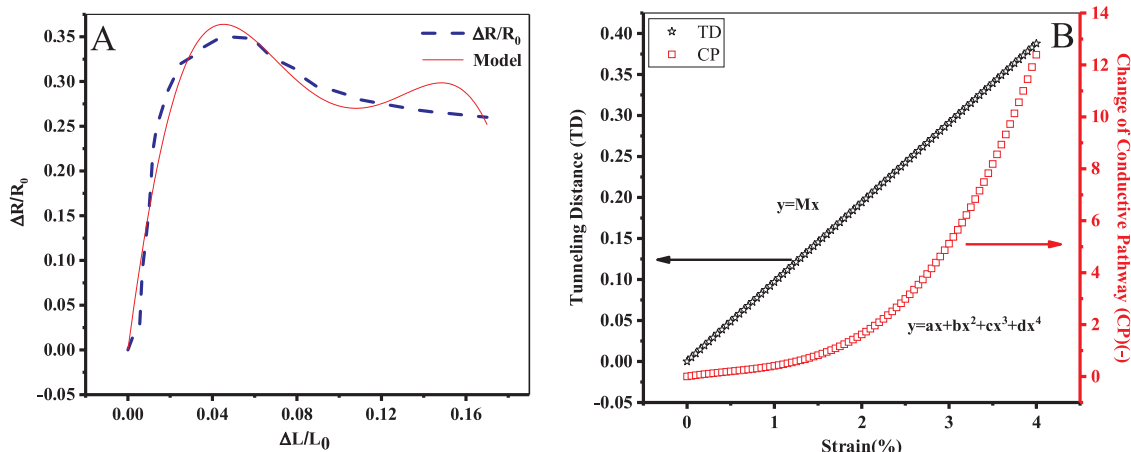


Fig. 6. (A) Theoretical and experimental results for $\Delta R/R_0$ against the applied strain, (B) The variation of TD and CP versus strain in the linear range, for BC/MWCNTs nanocomposite aerogel.

upon cyclic loading/unloading. A gauge factor of 21 obtained for BC/MWCNTs nanocomposite aerogel, indicating the capability of nanocomposite aerogel as a strain sensor. Mathematical modeling results, based on tunneling theory, showed a proper fitting with experimental data. The practical application of BC/MWCNTs nanocomposite aerogel strain sensor in human detection motion evaluated. These findings reveal that the nanocomposite aerogel based on BC/MWCNT is a good candidate to construct a novel flexible and lightweight strain sensor.

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Appendix A. Supplementary data

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References

- Cao, X., Wei, X., Li, G., Hu, C., Dai, K., Guo, J., ... Guo, Z. (2017). Strain sensing behaviors of epoxy nanocomposites with carbon nanotubes under cyclic deformation. *Polymer*, 112, 1–9.
- Demilecamps, A., Beauger, C., Hildenbrand, C., Rigacci, A., & Budtova, T. (2015). Cellulose-silica aerogels. *Carbohydrate Polymers*, 122, 293–300.
- Erbaş Kiziltas, E., Kiziltas, A., Blumentritt, M., & Gardner, D. J. (2015). Biosynthesis of bacterial cellulose in the presence of different nanoparticles to create novel hybrid materials. *Carbohydrate Polymers*, 129, 148–155.
- He, H., Zhang, D., & Xu, X. (2012). Electrically conductive multiwall carbon nanotubes/poly(vinyl alcohol) composites with aligned porous morphologies. *Journal of Macromolecular Science Part B-Physics*, 51(12), 2493–2498.
- Hosseini, H., Kokabi, M., & Mousavi, S. M. (2018). BC/rGO conductive nanocomposite aerogel as a strain sensor. *Polymer*, 137, 82–96.
- Hu, C., Li, Z., Wang, Y., Gao, J., Dai, K., Zheng, G., ... Guo, Z. (2017). Comparative assessment of the strain-sensing behaviors of polylactic acid nanocomposites: Reduced graphene oxide or carbon nanotubes. *Journal of Materials Chemistry C*, 5(9), 2318–2328.
- Huang, Z.-M., Liu, X.-Y., Wu, W.-G., Li, Y.-Q., & Wang, H. (2017). Highly elastic and conductive graphene/carboxymethylcellulose aerogels for flexible strain-sensing materials. *Journal of Materials Science*, 52(20), 12540–12552.
- Ku-Herrera, J. J., Avilés, F., & Seidel, G. D. (2013). Self-sensing of elastic strain, matrix yielding and plasticity in multiwall carbon nanotube/vinyl ester composites. *Smart Materials & Structures*, 22(8), 85003.
- Liu, H., Li, Y., Dai, K., Zheng, G., Liu, C., Shen, C., ... Guo, Z. (2016). Electrically conductive thermoplastic elastomer nanocomposites at ultralow graphene loading levels for strain sensor applications. *Journal of Materials Chemistry C*, 4(1), 157–166.
- Liu, Y.-J., Cao, W.-T., Ma, M.-G., & Wan, P. (2017). Ultrasensitive wearable soft strain sensors of conductive, self-healing, and elastic hydrogels with synergistic “soft and hard” hybrid networks. *ACS Applied Materials & Interfaces*, 9(30), 25559–25570.
- Liu, H., Jiang, H., Du, F., Zhang, D., Li, Z., & Zhou, H. (2017). Flexible and degradable paper-based strain sensor with low cost. *ACS Sustainable Chemistry & Engineering*, 5(11), 10538–10543.
- Lv, P., Feng, Q., Wang, Q., Li, G., Li, D., & Wei, Q. (2016). Biosynthesis of bacterial cellulose/carboxylic multi-walled carbon nanotubes for enzymatic biofuel cell application. *Materials*, 9(3), 183.
- Minjeong, H., Seongdong, L., Jonghwa, P., Doo-Seung, U., Youngoh, L., & Hyunhyub, K. (2015). Bioinspired interlocked and hierarchical design of ZnO nanowire arrays for static and dynamic pressure-sensitive electronic skins. *Advanced Functional Materials*, 25(19), 2841–2849.
- Olsson, R. T., Azizi Samir, M. A. S., Salazar-Alvarez, G., Belova, L., Ström, V., Berglund, L. A., ... Gedde, U. W. (2010). Making flexible magnetic aerogels and stiff magnetic nanopaper using cellulose nanofibrils as templates. *Nature Nanotechnology*, 5(8), 584.
- Park, S., Park, J., Jo, I., Cho, S. P., Sung, D., Ryu, S., ... Kim, B. S. (2015). In situ hybridization of carbon nanotubes with bacterial cellulose for three-dimensional hybrid bioscaffolds. *Biomaterials*, 58, 93–102.
- Park, W.-I., Kim, H.-S., Kwon, S.-M., Hong, Y.-H., & Jin, H.-J. (2009). Synthesis of bacterial celluloses in multiwalled carbon nanotube-dispersed medium. *Carbohydrate Polymers*, 77(3), 457–463.
- Ramuz, M., Tee, B. C.-K., Tok, J. B.-H., & Bao, Z. (2012). Transparent, optical, pressure-sensitive artificial skin for large-area stretchable electronics. *Advanced Materials*, 24(24), 3223–3227.
- Sadeghi, S., Arjmand, M., Otero Navas, I., Zehtab Yazdi, A., & Sundararaj, U. (2017). Effect of nanofiller geometry on network formation in polymeric nanocomposites: comparison of rheological and electrical properties of multiwalled carbon nanotube and graphene nanoribbon. *Macromolecules*, 50(10), 3954–3967.
- Sun, K., Xie, P., Wang, Z., Su, T., Shao, Q., Ryu, J., ... Guo, Z. (2017). Flexible polydimethylsiloxane/multi-walled carbon nanotubes membranous metacomposites with negative permittivity. *Polymer*, 125, 50–57.
- Ul-Islam, M., Khan, T., & Park, J. K. (2012). Water holding and release properties of bacterial cellulose obtained by in situ and ex situ modification. *Carbohydrate Polymers*, 88(2), 596–603.
- Wan, C., & Li, J. (2015). Facile synthesis of well-dispersed superparamagnetic $\gamma\text{-Fe}_2\text{O}_3$ nanoparticles encapsulated in three-dimensional architectures of cellulose aerogels and their applications for Cr(VI) removal from contaminated water. *ACS Sustainable Chemistry & Engineering*, 3(9), 2142–2152.
- Wang, Y., Weng, G. J., Meguid, S. A., & Hamouda, A. M. (2014). A continuum model with a percolation threshold and tunneling-assisted interfacial conductivity for carbon nanotube-based nanocomposites. *Journal of Applied Physics*, 115(19), 193706.
- Wang, Z., Qi, J., Yan, X., Zhang, Q., Wang, Q., Lu, S., ... Zhang, Y. (2013). A self-powered strain sensor based on a ZnO/PEDOT: PSS hybrid structure. *RSC Advances*, 3(38), 17011–17015.
- Yan, Z., Chen, S., Wang, H., Wang, B., & Jiang, J. (2008). Biosynthesis of bacterial cellulose/multi-walled carbon nanotubes in agitated culture. *Carbohydrate Polymers*, 74(3), 659–665.
- Yan, Z., Chen, S., Wang, H., Wang, B., Wang, C., & Jiang, J. (2008). Cellulose synthesized by *Acetobacter xylinum* in the presence of multi-walled carbon nanotubes. *Carbohydrate Research*, 343(1), 73–80.
- Yao, Q., Fan, B., Xiong, Y., Wang, C., Wang, H., Jin, C., ... Sun, Q. (2017). Stress sensitive electricity based on Ag/cellulose nanofiber aerogel for self-reporting. *Carbohydrate Polymers*, 168, 265–273.
- Yoon, S. H., Jin, H.-J., Kook, M.-C., & Pyun, Y. R. (2006). Electrically conductive bacterial cellulose by incorporation of carbon nanotubes. *Biomacromolecules*, 7(4), 1280–1284.
- Zhang, Z., Zhang, J., Zhao, X., & Yang, F. (2015). Core-sheath structured porous carbon nanofiber composite anode material derived from bacterial cellulose/polypyrrole as an anode for sodium-ion batteries. *Carbon*, 95, 552–559.
- Zhao, J., Dai, K., Liu, C., Zheng, G., Wang, B., Liu, C., ... Shen, C. (2013). A comparison between strain sensing behaviors of carbon black/polypropylene and carbon nanotubes/polypropylene electrically conductive composites. *Composites Part A, Applied Science and Manufacturing*, 48, 129–136.
- Zheng, Y., Li, Y., Dai, K., Liu, M., Zhou, K., Zheng, G., ... Shen, C. (2017). Conductive thermoplastic polyurethane composites with tunable piezoresistivity by modulating the filler dimensionality for flexible strain sensors. *Composites Part A, Applied Science and Manufacturing*, 101, 41–49.
- Zhi, X., Zhang, H., Bin, Liao, Y. F., Hu, Q. H., Gui, C. X., ... Yu, Z. Z. (2015). Electrically conductive polycarbonate/carbon nanotube composites toughened with micron-scale voids. *Carbon*, 82, 195–204.
- Zhuo, H., Hu, Y., Tong, X., Chen, Z., Zhong, L., Lai, H., ... Sun, R. (2018). A super-compressible, elastic, and bendable carbon aerogel with ultrasensitive detection limits for compression strain, pressure, and bending angle. *Advanced Materials*, 30(18), 1706705.